Naive Bayes Classification

Machine Learning

B.Tech. Information Technology

1 Introduction

Naive Bayes is a **probabilistic classification algorithm** based on Bayes' theorem. It predicts the class of a new observation by calculating the probability of each possible class and selecting the class with the highest probability.

It is called "Naive" because it assumes that the features are conditionally independent of one another given the class.

Concept

Naive Bayes is especially useful when we have multiple features and want to determine which class is most likely for a new observation.

Common applications include:

- Spam email detection
- Sentiment analysis
- Document and news classification
- Text classification
- Medical diagnosis as a probabilistic classification task

2 Bayes' Theorem

Bayes' theorem describes how a probability is updated when new evidence is available.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Here:

- $P(A|B)$ = Posterior probability: probability of A given B
- $P(B|A)$ = Likelihood: probability of observing B given A
- $P(A)$ = Prior probability of A
- $P(B)$ = Evidence or marginal probability of B

Important

For classification, we are interested in finding the probability of a class given the observed features:

$$P(Class|Features)$$

The class having the highest posterior probability is selected as the prediction.

3 Naive Bayes Formula

Suppose a data point has features $X_1, X_2, \ldots, X_n$ and belongs to one of the classes C.

Using Bayes' theorem:

$$P(C|X_1, X_2, \ldots, X_n) = \frac{P(X_1, X_2, \ldots, X_n|C)P(C)}{P(X_1, X_2, \ldots, X_n)}$$

The denominator is the same for all classes, so for classification we compare:

$$P(C|X_1, X_2, \ldots, X_n) \propto P(C)P(X_1, X_2, \ldots, X_n|C)$$

The Naive Bayes assumption is:

$$P(X_1, X_2, \ldots, X_n|C) \approx \prod_{i=1}^{n} P(X_i|C)$$

Therefore:

$$P(C|X_1, X_2, \ldots, X_n) \propto P(C) \prod_{i=1}^{n} P(X_i|C)$$

Important

The word "Naive" refers to the conditional independence assumption:

$$P(X_1, X_2|C) \approx P(X_1|C)P(X_2|C)$$

The features do not have to be truly independent in the real world. Naive Bayes simply makes this simplifying assumption to make computation easier.

4 Important Probability Terms

Term	Meaning
Prior probability	Probability of a class before observing features
Likelihood / Conditional probability	Probability of a feature value given a class
Posterior probability	Probability of a class after observing the features
Evidence	Overall probability of the observed features
Class score	Unnormalized value used to compare classes

5 Naive Bayes Classification Algorithm

The general procedure is:

1. Calculate the prior probability of each class.
2. For each class, calculate the conditional probability of every feature value.
3. For the new observation, multiply the prior probability by all relevant conditional probabilities.
4. Calculate the score for every possible class.
5. Compare the class scores.
6. Assign the class having the highest score.

For two features:

$$Score(C) = P(C)P(X_1|C)P(X_2|C)$$

For three features:

$$Score(C) = P(C)P(X_1|C)P(X_2|C)P(X_3|C)$$

6 Worked Example: Will a Customer Buy a Laptop?

Consider a classification problem in which we want to predict whether a customer will buy a laptop based on their age and income.

Features:

- Age: Young / Old
- Income: High / Low

Classes:

- Buy Laptop = Yes
- Buy Laptop = No

6.1 Training Dataset

Customer	Age	Income	Buy Laptop
C1	Young	High	Yes
C2	Young	Low	No
C3	Young	High	Yes
C4	Old	High	Yes
C5	Old	High	No
C6	Old	Low	No
C7	Young	Low	No
C8	Old	High	Yes

We have 8 training records.
New customer:

$$Age = Young, \qquad Income = High$$

We need to predict whether the customer will buy the laptop.

6.2 Step 1: Calculate Prior Probabilities

There are 4 "Yes" records and 4 "No" records.

$$P(Yes) = \frac{4}{8} = 0.5$$

$$P(No) = \frac{4}{8} = 0.5$$

6.3 Step 2: Calculate Conditional Probabilities

For the class **Yes**, there are 4 records:

$$P(Young|Yes) = \frac{2}{4} = 0.5$$

$$P(High|Yes) = \frac{3}{4} = 0.75$$

For the class **No**, there are 4 records:

$$P(Young|No) = \frac{2}{4} = 0.5$$

$$P(High|No) = \frac{1}{4} = 0.25$$

6.4 Step 3: Calculate Class Scores

For **Yes**:

$$Score(Yes) = P(Yes)P(Young|Yes)P(High|Yes)$$

$$Score(Yes) = 0.5 \times 0.5 \times 0.75 = 0.1875$$

For **No**:

$$Score(No) = P(No)P(Young|No)P(High|No)$$

$$Score(No) = 0.5 \times 0.5 \times 0.25 = 0.0625$$

6.5 Step 4: Compare the Scores

$$0.1875 > 0.0625$$

Therefore:

$$\boxed{Prediction = Yes}$$

Example

The Naive Bayes classifier predicts that the new customer is likely to **buy the laptop** because the score for the "Yes" class is higher than the score for the "No" class.

7 Why Are the Probabilities Multiplied?

Suppose there are two features: X_1 and X_2.

The Naive Bayes assumption allows us to approximate their joint probability as:

$$P(X_1, X_2|C) \approx P(X_1|C)P(X_2|C)$$

Therefore:

$$P(C|X_1, X_2) \propto P(C)P(X_1|C)P(X_2|C)$$

This makes the calculation simple even when there are many features.

Important

In a basic classification numerical, it is usually sufficient to calculate the class scores and compare them. The denominator $P(X_1, X_2)$ is common to all classes, so it does not affect which class has the highest probability.

8 Zero-Frequency Problem

A major problem in Naive Bayes occurs when a feature value has never appeared in a particular class in the training data.

Suppose:

$$P(X_i = v|C) = \frac{0}{N} = 0$$

Since Naive Bayes multiplies probabilities, a single zero makes the entire class score zero:

$$Score(C) = P(C) \times 0 \times P(X_2|C) \times \cdots = 0$$

This can incorrectly eliminate a class from consideration simply because that feature value was not observed in the training dataset.

Important

Zero-frequency problem: If any conditional probability is zero, the complete Naive Bayes class score becomes zero.

9 Laplace Smoothing

Laplace smoothing is used to handle the zero-frequency problem.

Instead of using:

$$P(X_i = v|C) = \frac{Count(X_i = v, C)}{Count(C)}$$

we add 1 to the numerator and add the number of possible values k to the denominator:

$$P(X_i = v|C) = \frac{Count(X_i = v, C) + 1}{Count(C) + k}$$

where:

- $Count(X_i = v, C)$ = number of times feature value v occurs in class C
- $Count(C)$ = total number of records belonging to class C
- k = number of possible values of that feature

9.1 Simple Laplace Smoothing Example

Suppose there are 4 Spam emails, and the feature "Contains Free" has two possible values: Yes and No.

Assume "Contains Free = No" occurs 0 times among the Spam emails.

Without smoothing:

$$P(Free = No|Spam) = \frac{0}{4} = 0$$

Using Laplace smoothing, since the feature has two possible values, $k = 2$:

$$P(Free = No|Spam) = \frac{0+1}{4+2} = \frac{1}{6}$$

Therefore, the probability is no longer zero.

Important

For a binary feature, $k = 2$. For a categorical feature having three possible values, $k = 3$, and so on.

9.2 Why Laplace Smoothing Is Important

Laplace smoothing:

- prevents zero probabilities,
- prevents an entire class score from becoming zero,
- is particularly important for text classification, where unseen words or feature values are common.

10 Types of Naive Bayes Classifiers

Different versions of Naive Bayes are used depending on the type of input features.

Type	Feature Type	Typical Use
Gaussian Naive Bayes	Continuous numerical values	Measurements such as height, weight, temperature, etc.
Multinomial Naive Bayes	Discrete counts or frequencies	Text classification using word/-document counts
Bernoulli Naive Bayes	Binary or Boolean features	Presence/absence of words or other binary features
Categorical Naive Bayes	Categorical variables	Features such as color, type, category, etc.

10.1 Gaussian Naive Bayes

Gaussian Naive Bayes is used when features are continuous numerical variables. It assumes that the feature values within each class approximately follow a Gaussian (normal) distribution.

Examples:

- Height
- Weight
- Temperature
- Sensor measurements

10.2 Multinomial Naive Bayes

Multinomial Naive Bayes is commonly used for text classification, where features represent counts or frequencies of terms.

Examples:

- Spam detection
- News classification
- Document classification

10.3 Bernoulli Naive Bayes

Bernoulli Naive Bayes is suitable when features are binary.

For example:

$$Contains_Free = Yes/No$$

$$Contains_Link = Yes/No$$

It is useful when the presence or absence of a feature is more important than its count.

10.4 Categorical Naive Bayes

Categorical Naive Bayes is suitable for categorical features with a finite number of categories. Examples include:

- Weather = Sunny / Rainy / Cloudy
- Color = Red / Blue / Green
- Education = School / Graduate / Postgraduate

11 Advantages of Naive Bayes

- Simple and easy to understand.
- Fast to train and predict.
- Works well with high-dimensional data.
- Requires relatively little training data.
- Particularly effective for many text-classification problems.
- Can work with different types of features through different Naive Bayes variants.
- Produces probabilistic class estimates.

12 Limitations of Naive Bayes

- The conditional independence assumption is often unrealistic.
- Highly correlated features can violate the assumption.
- Zero-frequency problems can occur without smoothing.
- Probability estimates may not always be well calibrated, even when classification accuracy is good.
- It may not capture complex relationships between features.

Important

Naive Bayes can still perform very well even though its independence assumption is a simplification. The goal is not to perfectly model every relationship between features, but to classify effectively.

13 Applications of Naive Bayes

- **Spam Detection:** Classifying emails as Spam or Not Spam.
- **Sentiment Analysis:** Classifying reviews or social-media text as positive, negative, or neutral.
- **Document Classification:** Assigning documents to predefined categories.

- **News Classification:** Categorizing news articles into topics.
- **Text Filtering:** Filtering and organizing large collections of text.
- **Medical Classification:** Supporting classification tasks based on observed symptoms or measurements.

14 Practice Example: Spam Email Classification

Consider the following training dataset. The task is to classify a new email as **Spam** or **Not Spam**.

Features:

- Contains "Free" word: Yes / No
- Contains Link: Yes / No
- Sender Unknown: Yes / No

Email	Free	Link	Unknown Sender	Class
E1	Yes	Yes	Yes	Spam
E2	Yes	Yes	No	Spam
E3	Yes	No	Yes	Spam
E4	No	Yes	Yes	Spam
E5	No	No	No	Not Spam
E6	No	Yes	No	Not Spam
E7	Yes	No	No	Not Spam
E8	No	No	Yes	Not Spam

For the new email:

$$Free = Yes, \qquad Link = Yes, \qquad Unknown\ Sender = Yes$$

Student Task

Use the Naive Bayes algorithm to determine whether the new email is Spam or Not Spam.

1. Calculate $P(Spam)$ and $P(Not\ Spam)$.
2. Calculate all required conditional probabilities.
3. Calculate the class score for Spam.
4. Calculate the class score for Not Spam.
5. Compare the two scores.
6. Give the final prediction.
7. Check whether any zero-frequency problem occurs.
8. If a zero-frequency problem occurs, apply Laplace smoothing and repeat the calculation.

Important

Do not skip the probability table. Show the prior probabilities, conditional probabilities, class scores, and final comparison clearly in your solution.